

Requirements Analysis Document

Project: E-Commerce Platform

1. Project Description

Field	Details
Project Name	E-Commerce Platform
Objective	Build a user-friendly web-based platform for customers to browse, select, and purchase products online
Audience	End users (shoppers), Admin, Development & QA Teams

2. Stakeholders

Role	Responsibility
Product Owner	Requirement approval & backlog prioritization
QA Lead	Test strategy & quality standards
End Users (Shoppers)	Primary product consumers
Developer	Deliver and maintain features

3. Business Drivers

Driver	Description
Revenue	Functional cart and checkout flow directly enables sales

Driver	Description
User Trust	Stable login, correct product display, and ratings build user confidence
Brand Standards	EPAM-aligned header and UI ensures professional brand identity
Quality	SonarQube + CI/CD enforces code standards and reduces production bugs

4. Functional Requirements

4.1 User Authentication

ID	Requirement	Linked Issue
FR-001	Users shall be able to create a new account with name, email, and password	EPMICMPEKT-26
FR-002	Users shall be able to log in with registered credentials securely	—
FR-003	Users shall be able to reset their password via a registered email link	EPMICMPEKT-25
FR-004	Login page shall display the platform logo without errors	EPMICMPEKT-23

4.2 Product Catalog

ID	Requirement	Linked Issue
FR-005	System shall display products with images, descriptions, price, and ratings	EPMICMPEKT-28
FR-006	Product image shall update dynamically when a color variant is selected	EPMICMPEKT-29
FR-007	Product ratings shall render accurately based on available review data	EPMICMPEKT-28

4.3 Shopping Cart

ID	Requirement	Linked Issue
FR-008	Users shall be able to add products to cart from the Product Details Page	EPMICMPEKT-27
FR-009	Cart item count in the header shall update immediately on add/remove	EPMICMPEKT-27
FR-010	System shall display an appropriate message if a product is out of stock	—

4.4 Header & Navigation

ID	Requirement	Linked Issue
FR-011	Application header shall conform to the EPAM Standard Header design spec	EPMICMPEKT-22
FR-012	Notification icon shall respond to clicks and display user notifications	EPMICMPEKT-30

5. Non-Functional Requirements

ID	Category	Requirement
NFR-001	Performance	Product pages shall load within 3 seconds under normal conditions
NFR-002	Security	User credentials must be transmitted over HTTPS and stored encrypted
NFR-003	Code Quality	All code must pass SonarQube quality gate before merging
NFR-004	Test Coverage	Minimum 80% unit test coverage required on all new modules

ID	Category	Requirement
NFR-005	Maintainability	CI/CD pipeline must auto-run tests on every code push
NFR-006	Compliance	UI must conform to EPAM UX/UI design standards
NFR-007	Scalability	Architecture must support horizontal scaling
NFR-008	Reliability	All API endpoints must return appropriate error codes and messages

6. Technical Requirements

ID	Requirement
TR-001	CI/CD pipeline configured in Jenkins with unit test integration
TR-002	SonarQube configured and integrated into the pipeline as a quality gate
TR-003	Git repository set up with a defined branching strategy (Git Flow)
TR-004	Test automation framework set up and ready for use by Sprint 1
TR-005	REST API contracts defined and documented before Sprint 1

7. Assumptions

- 1. The platform is web-based — no native mobile app in scope
- 2. Jenkins is the agreed CI/CD tool
- 3. APIs are REST-based with JSON responses
- 4. Git Flow is the agreed branching model
- 5. Sprint 0 produces no user-facing features — setup and analysis only
- 6. Bugs listed above were found on an existing pre-built codebase

8. Risks & Dependencies

Item	Type	Impact	Mitigation
API contracts not finalized	Risk	High	Mock APIs; contract-first approach
CI/CD setup delayed	Risk	High	Prioritize DevOps stories in Sprint 0
SonarQube access/config	Dependency	High	Confirm infra access in Sprint 0
EPAM Header design spec	Dependency	Medium	Obtain spec before Sprint 1